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A Hybrid A*-APF Path Planning Method for Ships Entering the Berthing Waters

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ABSTRACT

Ship berthing is a critical and high-risk phase of navigation that requires highly precise path planning in environments with numerous obstacles. This paper presents a two-stage hybrid path planning approach designed to improve both safety and manoeuvrability during berthing operations. The first stage involves constructing an accurate environmental model based on berth characteristics. In the second stage, an enhanced A* algorithm is introduced with a directional consistency penalty to generate globally feasible paths with improved continuity. To further enhance local obstacle avoidan ce and edge-following capabilities, the artificial potential field method is applied. The resulting path is coupled with a ship dynamics model and a dynamic look-ahead strategy combined with PID control is employed to enable closed-loop heading and speed tracking. Simulation results show that the proposed method significantly enhances path smoothness and obstacle clearance. Specifically, the average heading change is reduced to 2.51° and the minimum obstacle distance increases from 15.12 to 39.04 m. This approach offers a practical and effective solution for autonomous berthing in constrained port environments.

1 | Introduction

Maritime transportation forms the backbone of global trade, with over 80% of goods transported by sea according to the International Maritime Organisation [1] and this proportion is steadily increasing [2]. Concurrently, maritime traffic density is also increasing, especially in high-traffic areas such as port entry channels and nearshore waters. These congested areas impose greater constraints on ship manoeuvring, raising significant concerns about navigational safety [3]. Ship navigation can be categorised into oceanic, coastal, nearshore and inland waterway navigation. Among these categories, coastal navigation is considered the most hazardous due to its complex and highly dynamic environmental conditions, which pose greater challenges to safe navigation [4]. The Yangtze River Delta region has the highest concentration of ports in China. Located in the country's economic heartland, this region accounts for approximately one-quarter of China's total economic output [5]. Historical data indicate that approximately 80% of maritime accidents are caused by human factors [6]. Moreover, adaptive and constraint-aware control strategies have been shown to maintain system stability under limited sensing and dynamic constraints [4], highlighting the importance of robust control methods when navigating complex and restricted environments such as berthing waters [7]. Therefore, improving the intelligence level of ship decision-making and control systems is a key strategy for reducing human error and enhancing navigational safety. Among the many challenges in this domain, ship path planning and trajectory tracking remain among the most critical.

Numerous methods have been proposed for ship path planning and tracking at different spatial and operational scales. Based on the scale of the navigation problem, these methods can generally be divided into global and local path planning. Global path

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planning includes algorithms such as A*, rapidly-exploring random tree (RRT), ant colony optimisation and genetic algorithms, which are used to generate feasible paths in known environments from start to destination [8]. Local path planning methods, such as the artificial potential field (APF) and the dynamic window approach (DWA), emphasise real-time performance, focusing on dynamic obstacle avoidance and short-term path adjustment [9]. The A* algorithm, as a classic heuristic search algorithm, has good path convergence and high efficiency performance. However, in complex environments, it often leads to issues such as path irregularity and excessive changes in heading due to dense nodes. Previous studies have attempted to improve the heuristic function of the A* algorithm, such as introducing a 'path length and collision risk' weighting mechanism [10]. While these approaches improve certain aspects, they typically rely on empirically designed cost functions lacking physical interpretability or smooth tracking capability. In contrast, the APF method excels in real-time obstacle avoidance and path smoothness. However, it suffers from limitations such as susceptibility to local minima and target inaccessibility [11].

Despite the progress made in global and local path planning, significant challenges remain for ship navigation in berthing waters. These environments are characterised by narrow channels, irregular coastlines, dense island distributions and strict manoeuvrability limitations, which make both global path generation and local obstacle avoidance substantially more complex than in open-sea scenarios. Existing A*-based methods often rely on grid maps with simplified boundaries, leading to non-smooth paths that are difficult to track using real ship dynamics. APF-based methods, while suitable for local refinement, suffer from sensitivity to parameter tuning and are prone to local minima, especially when applied to real geographic polygons extracted from electronic charts. To address these challenges, this study proposes a unified two-stage hybrid method that integrates, first, an improved A* algorithm developed by introducing a direction-consistency penalty, enabling globally feasible paths with enhanced smoothness. A local APF method is then used to refine path nodes, ensuring safe obstacle avoidance and effective edge-following [12]. In the second stage, the smoothed path is integrated with a three-degree-of-freedom (3-DOF) ship dynamics model, enabling coordinated control of speed and heading through PID controllers. This ensures full-process trajectory tracking under multiple constraints, including obstacle avoidance, safety margins and manoeuvring stability.

The structure of this paper is as follows: Section 2 presents a literature review on path planning and tracking methods; Section 3 introduces the ship motion model and the proposed planning and control strategy; Section 4 presents simulation-based case studies to evaluate the performance of the proposed method; and Section 5 concludes the study and outlines directions for future research.

2 | Literature Review

2.1 | Path Planning at Different Scales

Ship path planning is generally classified into two main categories based on navigation regions and mission characteristics: transoceanic path planning and coastal path planning.

Transoceanic path planning is typically used for long-distance voyages involving large merchant ships. It takes into account macro-level environmental factors such as ocean currents, wind, waves and as well as extreme weather conditions such as typhoons [13]. The key challenges lie in route optimisation under uncertainty, balancing voyage duration with fuel consumption and safety. Metaheuristic optimisation techniques, such as genetic algorithms, particle swarm optimisation and ant colony optimisation, are commonly employed, as they support multi-objective optimisation frameworks and can be coupled with weather or current forecasts. However, these methods tend to involve high computational complexity and are prone to local optima [14].

Wu et al. [15] developed a multi-scale visibility graph approach based on electronic navigational charts, which improves accuracy in representing coastlines while enhancing computational efficiency. Du et al. [16] introduced a novel method involving the creation of a new waypoint perpendicular to the great circle route and solved the optimal problem using dynamic programming. Specifically, this method was applied to simulate the initialisation path of a VLCC. Zhao et al. [17] proposed a hybrid multi-objective ship path planning method based on an improved particle swarm optimisation genetic algorithm.

By contrast, coastal path planning deals with short-distance navigation such as nearshore transit, autonomous berthing and obstacle avoidance [18]. These scenarios are characterised by dynamic environmental changes, dense obstacle distributions and greater demands for real-time adaptability and path precision. As a result, coastal path planning often combines graph search methods (e.g., A*, enhanced A*), APF and geometric modelling approaches such as Constrained Delaunay Triangulation. These are usually integrated with local path smoothing or dynamic obstacle avoidance algorithms to ensure real-world applicability.

Pan et al. [19] proposed an automatic route design algorithm for USV navigation, using Constrained Delaunay Triangulation for environmental modelling, with a focus on navigation safety and environmental impact. Grifoll et al. [20] developed ship meteorological navigation planning software using the A* algorithm and optimised ship routes based on wave effects.

In recent years, mixed methods and machine learning have introduced novel paradigms and methodological advancements to path planning in complex coastal environments. Accordingly, this study focuses on coastal path planning, using the typical complex waters of Zhoushan Port as the research area. It considers the topographical distribution characteristics of islands, reefs, water boundaries and berthing channels and combines A* and APF with tracking control strategies to design an integrated path planning and control method adapted to real-world scenarios.

2.2 | Path Planning Algorithms

Traditional global path planning algorithms such as A*, Dijkstra and RRT are capable of generating globally optimal paths in known environments [21]. However, they face limitations such as insufficient path smoothness and high computational cost

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TABLE 1 | Comparison of path planning methods.

Method	Advantages	Limitations	Literature
APF	Good real-time performance and computation efficiency; suitable for online path generation.	Prone to local minima; Paths are not smooth; Poor adaptability to complex environments.	Song et al. Tian et al. Liu et al.
A*/D*/Dijkstra	Strong path accessibility ensures globally optimal solutions.	High computational load; Not suitable for dynamic obstacle environments; path curvature is not smooth.	Singh et al. Guo et al. Hart et al.
RRT/RRT*	Performs well in high-dimensional complex environments and is suitable for dynamic scenes.	The path is highly random and not smooth enough, making it difficult to handle.	Zhang et al. Ye et al.

in high-resolution grids. Although A* and Dijkstra algorithms are widely used due to their reliability and convergence, the resulting paths often include sharp turns, which exceed a ship's minimum turning radius and reduce feasibility for tracking [22, 23].

To address these limitations, Singh et al. [24] proposed a new A* algorithm that is based on the classical A* algorithm. It incorporates specific constraints and environmental factors, such as dynamic obstacles and ocean currents. It also considers collision risks and navigational rules to find the optimal and safe path. Guo et al. [25] proposed a path planning algorithm that provides full coverage of the navigable area based on an improved A* algorithm. They used the grid method and the longest edge back-and-forth traversal algorithm to improve path planning efficiency, addressing the issues of low operational efficiency and high energy consumption in unmanned ship operations. Hart et al. [26] combined the A* algorithm with Dijkstra's shortest path method with a heuristic function to effectively reduce the search space while ensuring the optimal solution. Ye et al. [27] proposed the IKB-RRT algorithm, which enhances APF using rolling window technology to improve path planning efficiency and smoothness. Fan et al. [28] improved search efficiency by adopting a goal-biased strategy combined with a bidirectional RRT* algorithm. Naderolasli et al. [29]proposed a terminal sliding-mode disturbance observer-based adaptive neural controller for ASVs, which achieves rapid convergence and robust formation control under output constraints and external disturbances. Zhang et al. [30] proposed a flow-based velocity field RRT*(VF-RRT*) algorithm for path planning in ocean current fields defined by flow functions, which accelerates the convergence of upstream costs and generates smoother paths for USVs. While these methods have been applied to ship path planning, issues related to search speed and local optimisation still need further investigation [31]. Moreover, the integration of computer vision technologies has emerged as a promising direction to enhance environmental awareness and obstacle recognition. For example, Tang et al. [32] developed a vision-based obstacle avoidance system for mobile robots, which complements traditional planning methods by providing real-time perceptual inputs. Similarly, Hu et al. [33] utilised 3D vision for structural crack recognition, highlighting the potential of visual sensing in complex and dynamic environments. Despite these advancements, challenges such as local minima, slow convergence and path irregularity remain significant, particularly in complex coastal environments.

The APF is another widely used local path planning technique, inspired by the concept of potential energy in nature. It is favoured for its simplicity, real-time responsiveness and ability to produce relatively smooth and safe paths in dynamic environments. However, a well-known limitation of APF is its tendency to become trapped in local optima, known as the stagnation problem, causing the ship to fail in reaching its target. To address this issue, Tian et al. [34] proposed adjustments to the repulsive potential field by modifying the gravitational components and introducing a tuning factor, which helps reduce oscillations near obstacles and improves convergence. They also introduced a boundary potential field to constrain the navigable area and proposed an escape mechanism from local minima. Liu et al. [35] developed a two-potential-field fused adaptive path planning system, which integrates a velocity potential field for dynamic obstacles. This system adjusts the ship's path in real time based on velocity information, forming a comprehensive potential field when combined with the traditional APF. Song et al. [36] proposed a predictive APF method that incorporates time-domain information to generate smoother paths using anticipatory potentials.

Nevertheless, these improved APF approaches remain highly sensitive to parameter configurations. Complex gravitational and repulsive field configurations necessitate meticulous parameter calibration and improper parameters can negatively impact the accuracy and efficiency of ship path planning.

To illustrate the characteristics of mainstream algorithms, a comparative summary of mainstream algorithms is presented in Table 1.

2.3 | Hybrid Approaches and Integrated Path Tracking

To address the limitations of individual algorithms, several studies have proposed hybrid methods combining A* and APF. Chen et al. [37] integrated and improved the APF and A* algorithm to solve irregular obstacles in the forward direction. APF guides the mobile robot to accelerate forward motion and the A* algorithm with the three-neighbourhood search method enables precise obstacle avoidance. During obstacle avoidance, the current position and orientation vector of the mobile robot are constructed and the search range is restricted to within three neighbourhoods to avoid redundant searches. Naderolasli

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et al. [38] developed a fixed-time multilayer neural network-based leader–follower control scheme for ASVs, capable of handling limited sensor range and actuator saturation while maintaining precise formation. Liu et al. [39] combined the A* algorithm, APF method and least squares algorithm to address the planning of smooth paths for robots, thereby avoiding the planning algorithm being trapped in local minima. Sang et al. [40] generated a globally optimal path using an improved heuristic A* algorithm, then divided the optimal path into multiple sub-target points to escape local minima by switching target points. However, these methods have not been applied to complex scenarios and do not consider real ship dynamics models and obstacle avoidance is not based on real geographic information [41].

However, most of these studies either simplify the environment or neglect realistic ship dynamics. Obstacle avoidance is typically based on artificial maps, not real-world geographical information and the ship motion model is often neglected. Additionally, many studies treat path planning and path tracking as two separate tasks, resulting in suboptimal coordination between trajectory generation and control execution. For ships with limited manoeuvrability, this decoupling significantly impairs tracking accuracy [42]. To bridge this gap, Yuan et al. [43] proposed an event-triggered adaptive horizon model predictive control method to enhance the real-time performance and computational efficiency of path tracking. The results demonstrated that the proposed scheme can effectively reduce computational costs while ensuring control performance. Naderolasli et al. [44] presented a Type-2 fuzzy robust adaptive controller for a 1-DOF robotic manipulator, effectively maintaining stability under nonlinear uncertainties. Wang et al. [45] adopted the deep deterministic policy gradient algorithm, utilising a four-network structure to form the core RL framework. Simulation results demonstrated robust tracking performance. However, the pre-training process of RL networks is highly time-consuming and requires substantial empirical exploration. Another major category of control algorithms is model-based methods. While there has been some development in path tracking, most research on ship optimal control focuses solely on path planning or path tracking. Additionally, integrating different methods in practical applications remains unreliable and inefficient in practical integration [46]. Therefore, there remains a gap in integrating path planning and tracking into a robust control system.

Compared to existing improvements on A* methods and path tracking methods based on a combination of collision risk and path length heuristic functions, the two-stage hybrid path planning and tracking method proposed in this paper demonstrates enhanced capability in addressing complex obstacle distributions and ship dynamics constraints. By integrating global and local strategies and introducing dynamic constraints, it achieves intelligent control of the entire process from open-sea navigation to confined water manoeuvring. The innovations of this study mainly include: (1) using real-world geographic polygons to construct obstacle regions, thereby avoiding empirical risk models; (2) applying a spline interpolation algorithm to smooth the initial path and introducing angular velocity restriction and a look-ahead mechanism in the control strategy to avoid sharp turns, thereby enhancing the ship's tracking capability and stability in dynamic environments; and (3) integrating the ship's manoeuvring modelling group (MMG) with dynamic look-ahead control to achieve a high degree of integration between path planning and tracking control, better aligning with actual ship navigation requirements.

3 | A*-APF Path Planning Method

3.1 | Research Methods Overview

The traditional A* algorithm is capable of generating globally optimal paths. However, due to grid discretisation, the resulting paths often contain excessive nodes and abrupt angular transitions, which are unsuitable for ship path tracking in constrained environments. To address this, an improved A* algorithm is proposed to obtain a smoother and more navigable initial path.

1. Establish a realistic vectorised environment model: Extract real electronic map information based on OSMnx, with water area boundaries, island contours and obstacle areas. Shapely is used to construct obstacle polygons, forming a vectorised navigation environment and providing a realistic geographical context for path planning.
2. Path planning and optimisation: The improved A* algorithm is used for initial path search, followed by APF to correct the path so that it stays away from obstacle boundaries and meets safe navigation requirements. Finally, cubic spline smoothing is applied to adjust the curvature of the path and reduce discontinuous turns.
3. Path tracking control: Based on the MMG with 3-DOF, a path tracking controller is constructed. A dynamic forward distance adjustment mechanism and PID heading control strategy are used to allow the ship to follow the planned trajectory at a controlled speed and improve overall path tracking accuracy.

The framework of this study is shown in Figure 1:

3.2 | Ship Motion Model

The MMG model was proposed by the Japan Towing Tank Conference. The MMG model is typically formulated using 6-DOF equations to describe ship motion, including roll, pitch, yaw, surge, sway and heave. The core idea of MMG is to decompose the fluid forces acting on the ship according to their physical meanings and consider the mutual interference between these forces, ultimately constructing an accurate mathematical model of ship motion.

Based on the MMG model, we established the ship motion model shown in Figure 2, where o-$x_o y_o$ is the spatial fixed coordinate system and o-xy is the ship fixed coordinate system (ship motion coordinate system). When describing a ship's motion in water, it is typically described using six degrees of freedom. This study primarily considers the efficiency of path planning by focusing on three degrees of freedom with sway, surge and yaw. The expression of the lateral velocity v is as follows:

$$v = v_m + X_G \tag{1}$$

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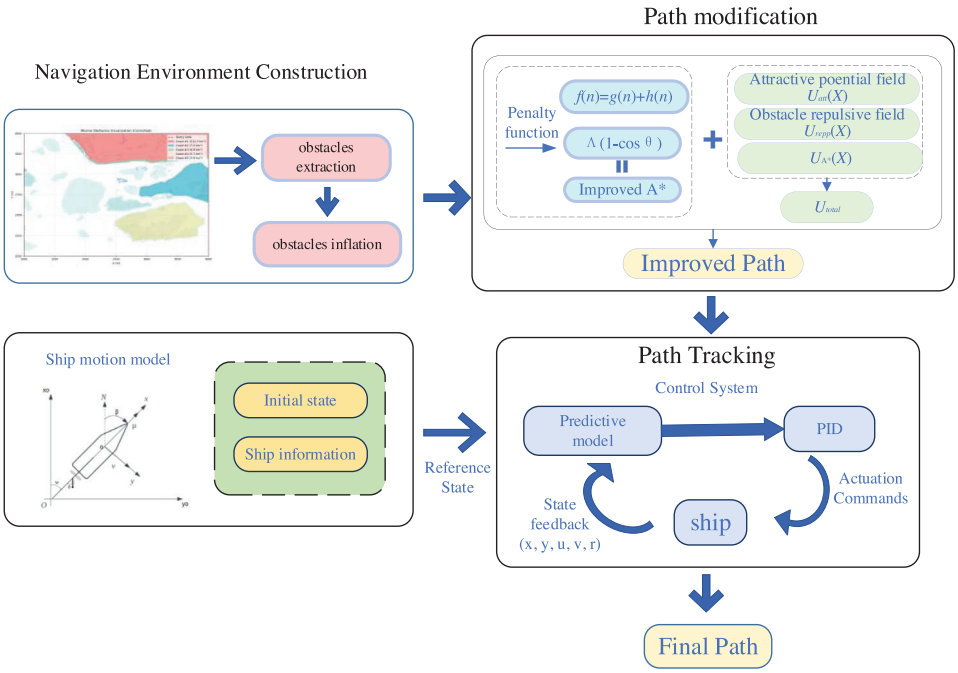

FIGURE 1 | Overall framework of the proposed path planning and tracking method.

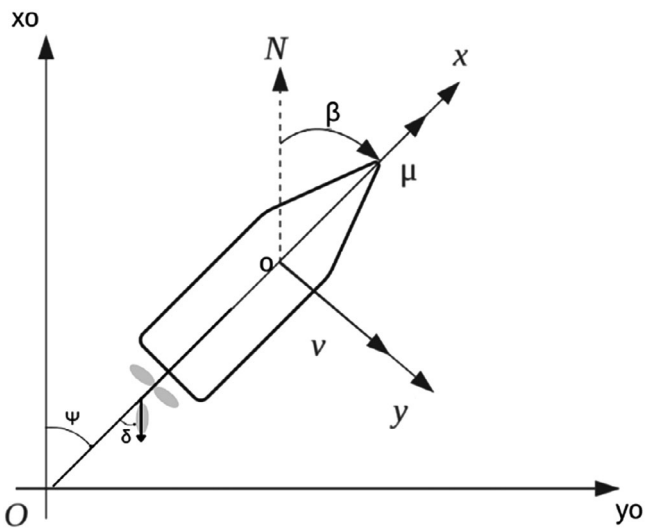

FIGURE 2 | Coordinate systems of ship motion.

where v is the lateral velocity of the ship (m/s), v_m represents the lateral velocity relative to the centre of mass, X_G denotes the correction term caused by the longitudinal shift of the centre of gravity (m).

The manoeuvring motion of a ship manifests as sway, surge and yaw. The motion equations are expressed as follows:

$$\begin{cases} m\ (\dot{u} - vr) = F_X \\ m\ (\dot{v} + ur) = F_y \\ L_{zG}\ \dot{r} = M_z \end{cases} \tag{2}$$

where u, v and r are the surge, sway and yaw velocities, m is the ship mass (kg), I_{zG} is the moment of inertia about the centre of gravity (kg·m^2), F_X, F_Y are the total hydrodynamic forces (N), M_Z is the yaw moment (N·m).

In Equation 2, the unknown variables are u, v and r. Here, the expressions for F_X, F_y, M_z are as follows:

$$\begin{cases} F_X = -m_x\dot{u} + m_y v_m r + X \\ F_Y = -m_y v_m - \dot{m}_x ur + Y \\ M_z = -J_z\dot{r} + N_m - x_G F_Y \end{cases} \tag{3}$$

where m_x, m_y are the added masses in surge and sway, J_z is the added moment of inertia in yaw, X, Y are the hydrodynamic forces of hull, rudder and propeller, N_m is the total yaw moment, x_G is the longitudinal distance between the origin and the centre of gravity (m).

Substitute Equations 1 and 3 into Equation 2 to eliminate v, resulting in Equation (4).

$$\begin{cases} (m + m_x)\,\dot{u} - (m + m_y)\,v_m r - x_G m r^2 = X \\ (m + m_y)\,\dot{v}_m - (m + m_x)\,ur + x_G m\dot{r} = Y \\ (I_{zG} + x_G^2 m + J_z)\,\dot{r} + x_G m\ (ur + \dot{v}_m) = N_m \end{cases} \tag{4}$$

Equation 4 is the set of motion equations to be solved. The X, Y and N_m on the right side of Equation 4 are defined as follows:

$$\begin{cases} X = X_H + X_R + X_P \\ Y = Y_H + Y_R \\ N_m = N_H + N_R \end{cases} \tag{5}$$

The subscripts H, R and P represent the hull, rudder and propeller, respectively.

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TABLE 2 | Controller and actuator parameters.

Parameter	Symbol	Value/range	Description
Max look-ahead	$L_{\max}$	180 m	Long-range stability
Min look-ahead	$L_{\min}$	40 m	Precision near target
Scaling coefficient	α	0.3	Dynamic scaling factor
PID (far)	K_p, K_i, K_d	2.0, 0.02, 0.5	Distant tracking
PID (near)	K_p, K_i, K_d	3.5, 0.05, 0.8	Fine control near goal
Rudder limit	$\delta_{\max}$	±35°	Physical actuator bound
Rudder rate	$\dot{\delta}_{\max}$	5°/s	Rate saturation
Propeller speed	$n_{\max}$	1.2 rev/s	Max propeller rotation

Table 2 summarises the key parameters used in the ship path tracking controller. It includes the look-ahead distances for dynamic preview control, PID gains for both far-field and near-goal tracking, as well as actuator limits, such as the maximum rudder angle, rudder rate and propeller speed. These parameters were selected based on a combination of ship manoeuvring characteristics, simulation tuning and physical actuator constraints. They ensure the controller achieves smooth and accurate trajectory tracking while respecting the physical limits of the vessel.

3.3 | Ship Path Planning

The proposed path planning approach comprises two key components: an improved A* algorithm for global path generation and spline-based smoothing that incorporates the ship's manoeuvrability constraints. This process ultimately yields a safe and navigable trajectory.

3.3.1 | Improved A* Algorithm for Path Optimisation

To generate feasible paths that meet the navigation requirements of actual ships, improvements were made to the traditional A* algorithm. Specifically, a direction consistency penalty term was introduced into the heuristic function to avoid frequent reversals and unnecessary turns, thereby improving overall smoothness and controllability. This algorithm is suitable for geographical environments containing real obstacle boundaries and can generate paths that more closely align with the ship's manoeuvring characteristics.

The core of the traditional A* algorithm lies in using a heuristic search strategy to find the lowest cost path from the starting point S to the target point G. Its evaluation function is expressed as:

$$f\ (n) = g\,(n) + h\,(n) \tag{6}$$

In this context, $f(n)$ denotes the total cost estimate for the current node n, $g(n)$ represents the actual cost from the starting point to the current node and $h(n)$ is the heuristic cost estimate from the current node to the goal point. In the traditional A* algorithm, Euclidean distance or Manhattan distance is typically used as $h(n)$. However, in a grid environment with diagonal movement capabilities, this study adopts an octile distance cost estimation, which better aligns with actual movement. It is defined as follows:

$$h_{\text{octile}}\ (n, G) = \Delta_{\max} + \sqrt{2}\Delta_{\min} \tag{7}$$

where $\Delta_{\max} = \max(|x_n - x_G|, |y_n - y_G|)$, $\Delta_{\min} = \min(|x_n - x_G|, |y_n - y_G|)$。

In a complex environment, the paths planned by traditional A* often contain many unnecessary zigzags and sharp directional changes, which are not conducive to execution for ships with dynamic constraints. To address this issue, this study introduces a direction consistency penalty term to enhance the 'directional continuity' of the path. The improved evaluation function is as follows:

$$f(n) = g\,(n) + h_{\text{octile}}\,(n, G) + \lambda\,(1 - cos\theta_n) \tag{8}$$

where θ_n represents the angle between the current node's forward direction and the ideal target direction: $cos\ \theta_n = \frac{(n-p)\cdot(G-n)}{\|n-p\|\cdot\|G-n\|}$;$p$ is the parent of the current node; λ is the direction penalty weight coefficient, empirically set to $\lambda = 80$ to amplify the cost of inconsistent directions. By introducing this penalty term, if the current expansion direction aligns with the direction toward the target (i.e., $\theta_n \approx 0$), the penalty is zero; when the path direction deviates significantly from the target direction, an additional cost is imposed to encourage the path to maintain overall directional consistency and reduce abrupt turns. The grid resolution was set to 15 m and each obstacle was inflated by 50 m to ensure safe navigable corridors around the island and land polygons.

These values were adjusted through repeated simulation to achieve stable convergence without introducing unnecessary detours.

As shown in Figure 3, the traditional A* algorithm typically allows expansion from the current node in eight directions, which are fixed in a grid map. The algorithm selects the next direction based on the cost function. However, the improved version proposed in this study adds a heading angle constraint to the traditional 8 directions, limiting the expansion direction to within ±35° of the current heading. This improvement aligns with the actual operations of ships, reducing large-angle turns and unreasonable movements. The improved path planning results in

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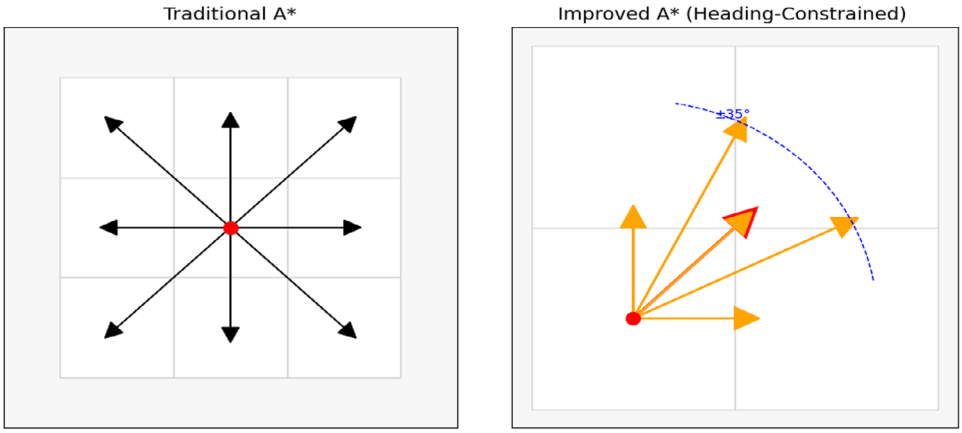
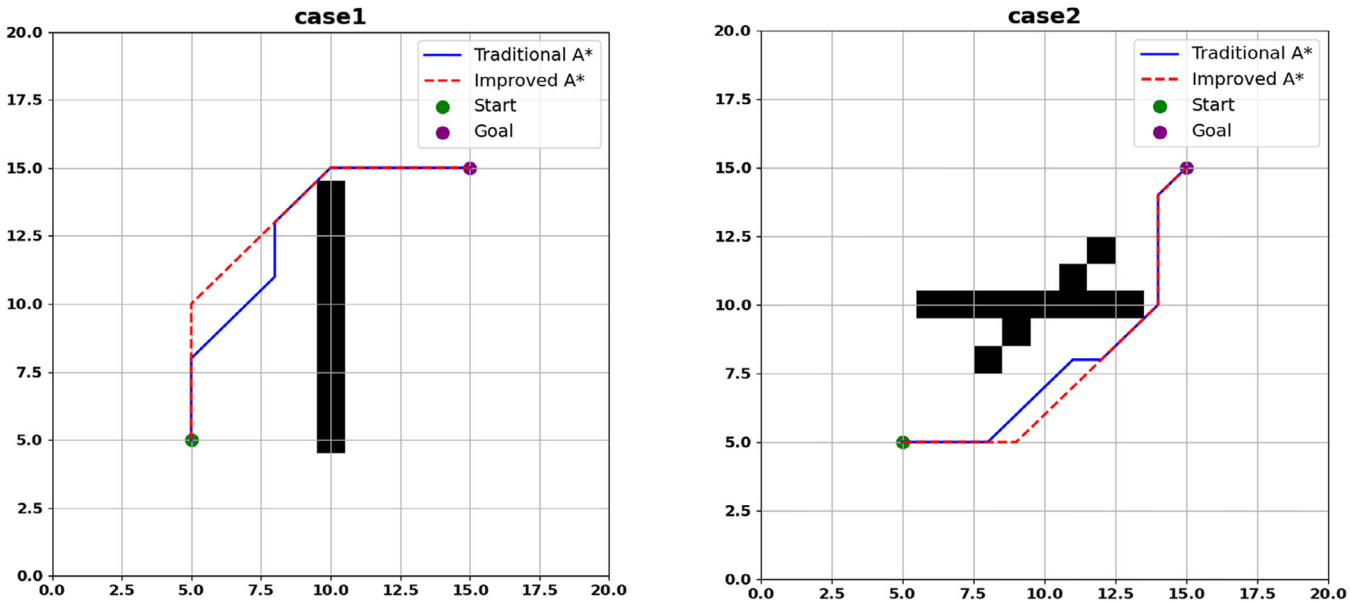


(a) Traditional A* (b) Improved A*

FIGURE 3 | Comparison of path search direction between traditional and improved A* algorithms.



(a) Obstacle layout 1 (b) Obstacle layout 2

FIGURE 4 | Path comparison between traditional and improved A* algorithms under different obstacles.

smoother paths that better match the movement characteristics, thereby reducing control difficulties during path tracking.

As shown in Figure 4, the traditional A* algorithm expands grid nodes primarily based on distance cost. It tends to expand nodes along directions that are geometrically closer to the goal, often reaching the edge of obstacle boundaries early and turning sharply along their contours. This results in paths with abrupt turns. In contrast, the improved A* algorithm introduces a heading-change penalty, which discourages sudden directional shifts during node expansion. As a result, the algorithm favours maintaining a consistent heading and generates smoother paths. For instance, when encountering an obstacle, the improved algorithm avoids closely skirting it, as the cost function accounts for additional penalties due to heading deviations. Although the heading-change penalty improves overall path smoothness, certain waypoints may still be located near obstacle boundaries, especially in complex environments. This occurs because the current improvement strategy focuses on suppressing sharp heading deviations rather than explicitly enforcing a minimum clearance from obstacles. In certain geometric configurations, maintaining a path close to the boundary may, in fact, represent the directionally optimal solution.

3.3.2 | APF Algorithm

The APF algorithm is widely used in local path planning due to its low computational complexity, rapid response and favourable path smoothness. In this study, a novel two-stage path planning framework is proposed by integrating the APF with the improved A* algorithm to enhance path safety and

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efficiency near obstacles. By incorporating the APF algorithm, additional costs are introduced for the A* algorithm within the obstacle influence range, enabling safe and efficient obstacle avoidance. Several modifications are applied to the conventional APF formulation to adapt the potential field to ship manoeuvring characteristics, particularly in berthing and constrained waters.

The total potential field U(X) is designed as the sum of three components: attraction, repulsion and a velocity-field weighted path attraction term:

$$U_{\text{total}}\ (X) = \sum_{j=1}^{N_p} \lambda_j \cdot \rho^2\left(X, P_j\right) + U_{\text{att}}(X) + U_{\text{repp}}(X) \tag{9}$$

where: $X = [x, y]^T$ is the ship's current position (m);x_g is the goal position (m); n is the number of obstacle polygons; N_p is the number of reference path points P_j; $\rho(X, P_j) = \| X - P_j \|$ is the Euclidean distance to the j-th path point (m);λ_j is the path weighting coefficient describing the attraction strength of the waypoint P_j.

Gravitational potential field: Always points toward the forward target point P_{tar} on the current path, driving the ship toward the goal path.

$$U_{\text{att}}(X) = \frac{1}{2} K_{\text{att}} \|X - X_g\|^2 \tag{10}$$

Repulsive potential field: Generated when the distance ρ between a ship and an obstacle is less than the threshold ρ_0, exerting a repulsive force to keep the ship away from hazardous regions.

$$U_{repp}(X) = \begin{cases} \frac{1}{2} K_{rep}\left(\frac{1}{\rho_j(X)} - \frac{1}{\rho_0}\right)^2, & \rho_j(X) \le \rho_0 \\ 0, & \rho\rho_0 \end{cases} \tag{11}$$

In this context, $\rho_j(X)$ is the minimum distance between the ship and the j-th obstacle (m) and $\rho_0 = 500$ m is the influence radius. The parameters $K_{\text{att}} = 1.5 \times 10^3$ and $K_{\text{rep}} = 8 \times 10^3$ correspond to the attraction and repulsion gains, respectively. The additional term $\sum_{j=1}^{N_p} \lambda_j \cdot \rho^2(X, P_j)$ introduces a velocity-field weighting effect, which ensures that the resultant guidance direction is consistent with the feasible manoeuvring direction of the MMG model.

The decay law for λ_j is defined as:

$$\lambda_j = \lambda_0\, e^{-\gamma d_j} \tag{12}$$

where $\lambda_0 = 1.0$, d_j is the distance from the ship to the waypoint P_j and $\gamma = 0.002$ controls the exponential decay rate.

The final combined force is:

$$F\ (X) = -\nabla U\ (X) = F_{\text{att}}\ (X) + F_{A*}(X) + F_{\text{rep}}(X) \tag{13}$$

By adding the target gravity, obstacle repulsion and path point gravity together, the gradient of the combined field can be calculated to determine the direction of the ship's movement. The resultant force $F(X)$ is the direction and trend in which the ship should move at this moment in the APF and the ship's target heading is the direction toward this force.

It is worth noting that the guidance direction generated by the APF is not directly used as a dynamic control command but rather as a reference heading input to the PID controller. This design ensures that the control actions comply with the ship's dynamic feasibility. In particular, the heading rate and turning radius produced by the APF are limited by the MMG model's kinematic constraints and the rudder saturation defined in Section 3.4. Therefore, although the APF provides instantaneous gradient directions, the actual manoeuvring path strictly follows the dynamic constraints, ensuring physical consistency among the A*, APF and MMG modules.

3.4 | Path Tracking Control

Once a smooth path is generated, a dynamic PID-based control strategy is employed to track the path. The control system includes dynamic look-ahead target selection, heading adjustment and speed modulation.

3.4.1 | Target Point Selection

Let the current ship position be defined as:

$$X\ (t) = [x(t), y(t)]^T, \tag{14}$$

$X(t)$ represents the two-dimensional spatial position of the ship at time t, which is a column vector. $x(t)$, $y(t)$ are the x-axis and y-axis coordinates of the ship in the plane coordinate system at time t, which change dynamically with time.

The global smooth path discrete point sequence is $\{P_k\}_{k=0}^N$. The index of the path point closest to the ship is computed as:

$$K_{\text{near}} = \underset{k}{\text{argmin}} \|X(t) - P_k\| \tag{15}$$

K_{near} is the closest path point to the current ship position in a sequence of discrete path points. $\|X(t) - P_k\|$ denotes the straight-line distance between the current ship position and the k-th path point. To ensure continuous path tracking, the ship's closest point on the discrete path is selected as the initial reference for computing the forward-looking target point.

Then, introduce the dynamic forward distance $L(t)$ (decreasing with the distance to the destination):

$$L\ (t) = \begin{cases} L_{\max},\ d_g(t) > d_{th} \\ \max\left(L_{\min}, \alpha d_g(t)\right), d_g(t) < d_{th} \end{cases} \tag{16}$$

where $d_g\ (t) = \|X(t) - P_k\|$, $\alpha \in (0, 1)$. $L(t)$: the dynamic look-ahead distance at time t (i.e., the length of the target path for the ship's current position looking forward).

This dynamic adjustment accounts for the ship's manoeuvring characteristics. When the ship is far away from the end point, the maximum look-ahead distance $L_{\max}$ is adopted to anticipate the overall trajectory and avoid sudden sharp turns; when the ship is close to the end point, the look-ahead distance is shortened with the distance to ensure accurate tracking and realise smooth berthing.

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3.4.2 | Heading Controller Settings

Based on the error between the current ship heading $\psi(t)$ and the target heading $\psi_d(t)$, a PID controller is used to generate a rudder angle command.

$$\psi_d\ (t) = \arctan2\left(x'_{\text{tar}} - x, y'_{\text{tar}} - y\right) \tag{17}$$

$\psi_d(t)$ denotes the target heading angle (i.e., the desired orientation of the ship). $x'_{\text{tar}}, y'_{\text{tar}}$ is the x and y coordinates of the forward-looking target point P. arctan2(Δy, Δx) is a four-quadrant arctangent function, which calculates the heading angle by $\Delta y = y'_{\text{tar}} - y$ and $\Delta x = x'_{\text{tar}} - x$.

The tracking error is expressed as follows:

$$e_\psi\ (t) = \text{wrap}\left(\psi_d\left(t\right) - \psi\left(t\right)\right) \tag{18}$$

$e_\psi(t)$ is the heading error at time t, the deviation of the target heading from the actual heading. Correcting angular errors by means of a winding function avoids false errors due to angular periodicity and ensures that subsequent PID controllers respond accurately to true deviations. The use of the arctan2 function in Equation (16) ensures a well-defined desired heading for all relative positions to the target point. Furthermore, the error wrapping in Equation (17) maintains the heading error within a continuous circular interval, preventing numerical issues.

Therefore, the rudder angle output is as follows:

$$u\ (t) = k_p\, e\left(t\right) + k_i \int_0^t e(t)dt + k_d \frac{de\left(t\right)}{dt} \tag{19}$$

Controller parameters are automatically switched based on the distance to the destination to achieve a quick response at the far end and stable docking at the near end.

The ship's propulsion speed η_{ps} uses a linear deceleration method.

$$\eta_{ps}\ (t) = \begin{cases} n_{\max},\ d\left(t\right) > d_{\text{slow}} \\ \max\left(n_{\min}, k_v \cdot d\left(t\right)\right), d\left(t\right) \le d_{\text{slow}} \end{cases} \tag{20}$$

$d(t)$ represents the Euclidean distance between the current ship and the target point. This formulation enables dynamic adjustment of propulsion speed, allowing the ship to maintain cruising velocity when the ship is far from the destination and gradually slow down as it approaches the destination.

4 | Case Studies

4.1 | Scenario 1: Rule-Based Target Obstacle Path

To verify the effectiveness of the proposed path planning and tracking method, a virtual test environment is constructed in which the ship navigates from a given starting point to a target location in a complex waterway with multiple obstacles. Seven static circular obstacles are manually configured to simulate typical maritime hazards such as reefs, buoy zones and restricted areas. The coordinates and radii of these obstacles are summarised in Table 3.

TABLE 3 | Coordinates and radii of static obstacles.

Number	Center coordinates (x, y)	Radii (m)
1	(2400, 1800)	100
2	(3100, 2700)	100
3	(2100, 1300)	140
4	(1900, 1500)	130
5	(2200, 2100)	150
6	(2800, 2700)	120
7	(2800, 2300)	140

In addition, to ensure a safe distance between the ship and obstacles, each obstacle is assigned a 15 m expansion zone as a safety buffer during the planning process. The initial path is generated by an improved A* algorithm. This algorithm introduces an obstacle distance penalty mechanism based on the traditional A* algorithm, accounting for the distance between each path point and its nearest obstacle. When the distance falls below 80 meters, a cost term is introduced with a penalty factor $\lambda = 80$. The attraction coefficient is set to $katt = 2.0$ and $Krep = 8.0$.

Figure 5 compares trajectories generated by three methods: (i) the global path from the traditional A* algorithm, (ii) the *global path from the improved A algorithm** and (iii) the final executed trajectory after APF refinement and smoothing. The three trajectories are evaluated under two test cases with different obstacle configurations. As shown, the improved method produces smoother, more navigable S-shaped paths while maintaining a greater distance from obstacles.

As shown in Table 4, 'Improved A*' refers to the A* algorithm with the directional consistency penalty only, while 'A* + APF' integrates the improved A* global path with APF-based local refinement and spline smoothing for tracking. The proposed method reduces the average turning angle from 4.17° (traditional A*) to 2.51° and increases the minimum distance to obstacles from 15.12 to 39.04 m. Although the total trajectory length increases slightly, this trade-off is justified by the substantial improvement in navigational safety and tracking stability.

4.2 | Scenario 2: Real-World Path

4.2.1 | Ship Model

The ship model used in this study is a scaled-down model of the KVLCC2-type VLCC oil tanker proposed by the National Maritime Research Institute of Japan. This model is widely used in manoeuvrability tests and MMG model simulation studies and it has publicly available, complete hydrodynamic parameters and widely recognised experimental data. The parameters of the ship model are shown in Table 5.

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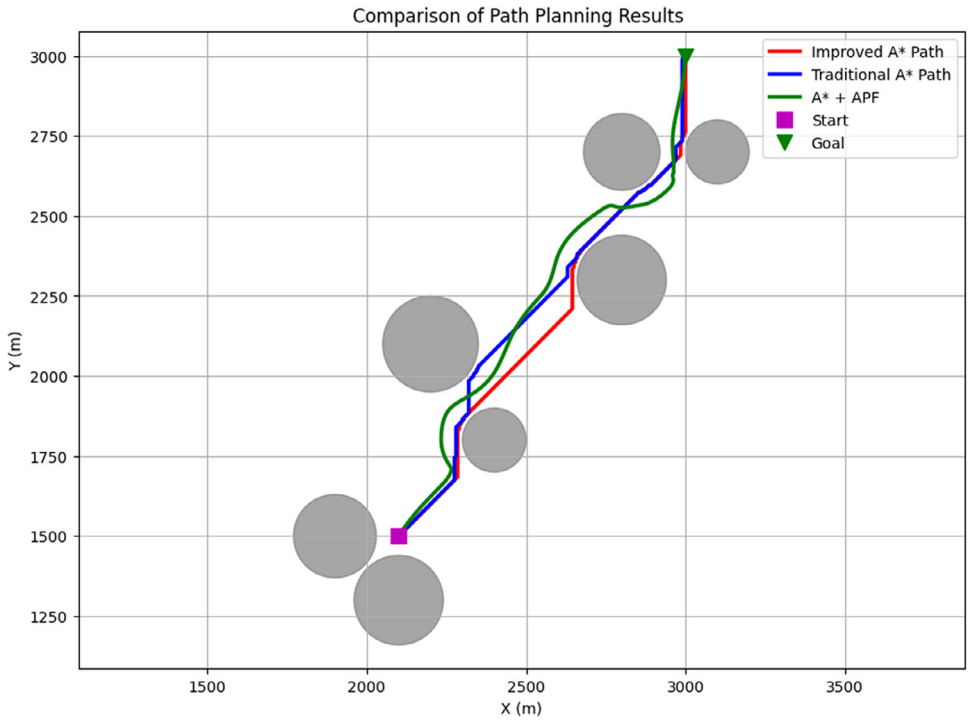

FIGURE 5 | Comparison of path planning results.

TABLE 4 | Comparison of performance parameters of different algorithms.

Indicator	Traditional A*	Improved A*	A* + APF
Total trajectory length (m)	1881.58	1985.88	1949.72
Minimum distance to obstacles (m)	15.12	37.14	39.04
Average turning angle (°)	4.17	10.44	2.51

TABLE 5 | Ship model parameters.

Parameters	Symbol	Value
Ship length/m	L	64.0
Ship width/m	B	11.6
Draft/m	d	4.16
Displacement/t	∇	2000.64
Rotational inertia/kg·m^2	Iz	Approximately scaled

This simulation uses the MMG with 3-DOF, with the model ship as the control object. The state variables are longitudinal velocity u, lateral velocity v and heading angle velocity r. The input control variables are propeller speed and rudder angle δ. The initial simulation conditions are as follows: initial heading angle $\psi_0 = 0$, initial velocity $X_0 = [3.85, 0.0, 0.0]$.

4.2.2 | Map Construction

To further assess the real-world applicability of the proposed method, a simulation is performed in the navigational waters surrounding Zhoushan Port, China. The map is constructed using real geographic data based on latitude and longitude coordinates. The selected region extends from 122.1°E to 122.5°E in longitude and from 29.8°N to 30.2°N in latitude, encompassing multiple bays, islands and navigation channels.

Figure 6a shows an electronic chart of the area, where latitude and longitude are converted to meter-level plane coordinates by Mercator projection and a local coordinate system is constructed. After the conversion, all obstacle points, starting and ending points are uniformly expressed in local plane coordinates (meters), as shown in Figure 6b.

To adapt the geographic data for simulation, latitude–longitude coordinates are first projected into a 2D Cartesian plane via the Mercator projection. The resulting coordinates are then uniformly scaled by a factor of 1:25 to ensure consistency between the physical environment and the simulation grid resolution. All start points, waypoints and obstacle polygons are subsequently transformed into the local coordinate system according to the following mapping rule:

$$\begin{cases} x_{\min} = S \cdot (x_{\text{utm}} - x_0) \\ y_{\min} = S \cdot (y_{\text{utm}} - y_0) \end{cases} \tag{21}$$

where S denotes the Scaling ratio, which is set to 0.04, with $(x_0,\ y_0)$ as the defined origin. Initial position $(x_0,\ y_0) = (2000.0, 1000.0)$. Endpoint position $(x_{\text{goal}}, y_{\text{goal}}) = (3500.0, 3200.0)$. All obstacle polygon boundary points are uniformly scaled and translated in accordance with these parameters.

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(a)Nautical chart for a portion of the Zhoushan Port

(b) Visualization of marine obstacles in the same area

FIGURE 6 | Part of Zhoushan port.

4.2.3 | Ship Path Tracking Control

The MMG model with 3 DOF was employed to simulate the ship's response during real-time path tracking. The same ship model and control parameters as in 4.2.1 were applied.

As shown in Figure 7, the traditional A* algorithm generated paths with frequent heading reversals, particularly in areas with densely distributed obstacles. This is due to its limited ability to handle complex spatial constraints, often resulting in redundant detours. This was attributed to its limited capacity to manage complex spatial constraints, often leading to redundant detours. In contrast, the improved A* algorithm reduced the number of nodes and turning points. Further enhancement was achieved by integrating the APF method, which incorporated real-time obstacle data and virtual forces to dynamically adjust the tracking direction.

Figure 8 presents the ship's speed and deviation angle over time. In Figure 8b, the ship's speed remained stable at approximately 7.2 kn (≈3.70 m/s), with only minor fluctuations in the initial stage and good stability during the system's start-up. The deceleration phase showed a continuous linear decrease without oscillations, demonstrating a smooth and effective speed control strategy. The final speed reached zero, confirming that the controller executed the stop command precisely near the target point. In Figure 8a, the deviation angle remained within ±10°, with a rapid initial response and final convergence, satisfying ship trajectory control requirements.

Figure 9a compares heading angle responses under the traditional and improved control strategies. The traditional method (green dashed line) exhibited pronounced oscillations and frequent large-amplitude fluctuations, indicating unstable heading control and a higher tendency for overshoot. The improved method (blue solid line) achieved a smoother heading trajectory, with reduced peak amplitudes and fewer fluctuations. After approximately 200 s, the heading angle under the improved method gradually converged and stabilised around the target value, whereas the traditional method continued to oscillate. These results verify that the improved strategy significantly enhanced heading control performance by suppressing overshoot and enabling smoother, more accurate trajectory tracking.

The mid-course fluctuations in Figure 9b were mainly caused by temporary rudder overshoot when the look-ahead point rapidly shifted at local curvature transitions. This transient deviation reflects the controller's fast response to direction changes and was subsequently damped by the dynamic PID gain scheduling. In Figure 9b, slightly larger fluctuations in the mid-course segment were observed, likely due to rudder overshoot caused by look-ahead point deviations or abrupt path curvature changes. Thereafter, the heading angle stabilised with only minor oscillations, indicating steady-state tracking. The rudder angle remained within ±15°, demonstrating that heading errors were progressively reduced and the system was stabilising. In the later stage, both heading and rudder angles converged, confirming the system's strong steady-state tracking capability.

To evaluate the robustness of our proposed path planning and tracking method, simulations were performed from multiple initial positions and headings. The initial conditions for the five trajectories are as follows:

1. Start: (2000, 1000), heading: 0°
2. Start: (2200, 1200), heading: 30°
3. Start: (1800, 1100), heading: -45°
4. Start: (2100, 900), heading: 60°
5. Start: (1900, 1300), heading: -30°

Simulations were conducted from multiple initial positions and headings, as reported in the study. For each configuration, key performance metrics — including path length, minimum clearance, average curvature, RMS cross-track error (XTE), maximum XTE and final docking error — were recorded. The results across the five initial configurations are summarised below (See Table 6):

These results indicate that our path planning and tracking approach produces consistent performance across different starting positions and headings. The variation in minimum clearance and cross-track error is within acceptable operational limits, demonstrating the robustness of the proposed method.

The resulting trajectories are shown in Figure 10. Each trajectory successfully reaches the same goal point, marked as a red star, while avoiding obstacles (grey areas). The planned reference path is plotted as a dashed line for comparison.

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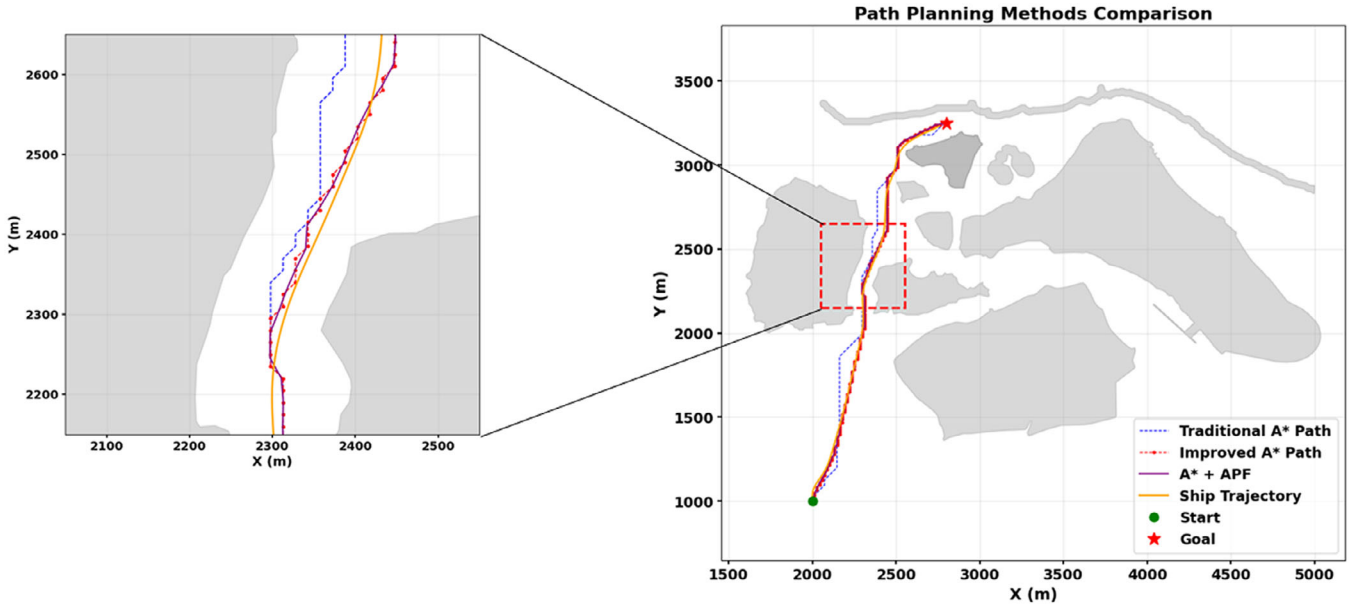
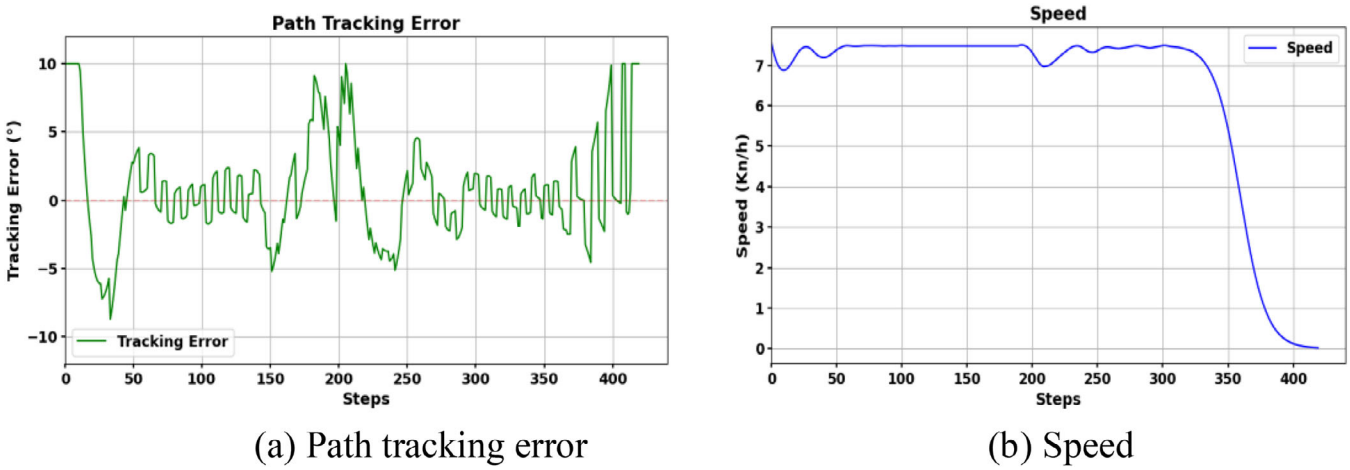


FIGURE 7 | Path tracking results in the real-world scenario.



(a) Path tracking error (b) Speed

FIGURE 8 | Ship tracking performance.

From Figure 10, it can be observed that, despite the differences in starting positions and headings, all trajectories converge smoothly toward the goal, demonstrating the consistency of the proposed method. Trajectories initialised farther from the main channel (e.g., Trajectories 3 and 5) show slightly larger initial offsets but realign with the reference path within a short distance. All resulting paths remain smooth, with comparable curvature and minimal cross-track deviation, indicating that the planner is robust to variations in initial heading.

4.2.4 | Methods Comparison

In trajectory tracking tasks, traditional path planning algorithms, such as the A* algorithm, could generate globally feasible paths; however, they were prone to issues such as path redundancy and limited flexibility in obstacle avoidance when operating in dynamic environments or under locally complex constraints. The APF method simulated attractive and repulsive potential fields to locally regulate global paths, thereby improving trajectory smoothness and the rationality of obstacle avoidance. Experiments using three-dimensional (3D) potential field visualisation and projection analysis validated the performance improvements achieved by integrating APF into trajectory tracking.

As shown in Figure 11a, there was a significant difference between the traditional A* path (red) and the improved path (green) within the attractive potential field. The traditional A* path exhibited noticeable detours in medium-to-high potential field regions, relying solely on discrete node search. In contrast, the improved path followed low-potential field channels, leveraging real-time sensing of potential field gradients to achieve smoother navigation while seeking shorter, more efficient routes. The potential field visualization demonstrated that the improved path aligned more closely with the optimal gradient trajectory, thereby reducing travel distance and energy consumption.

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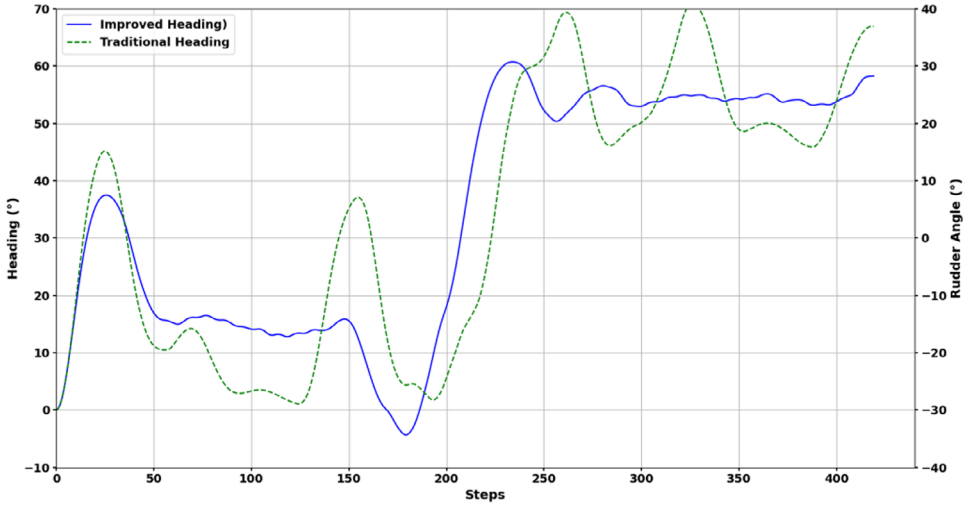
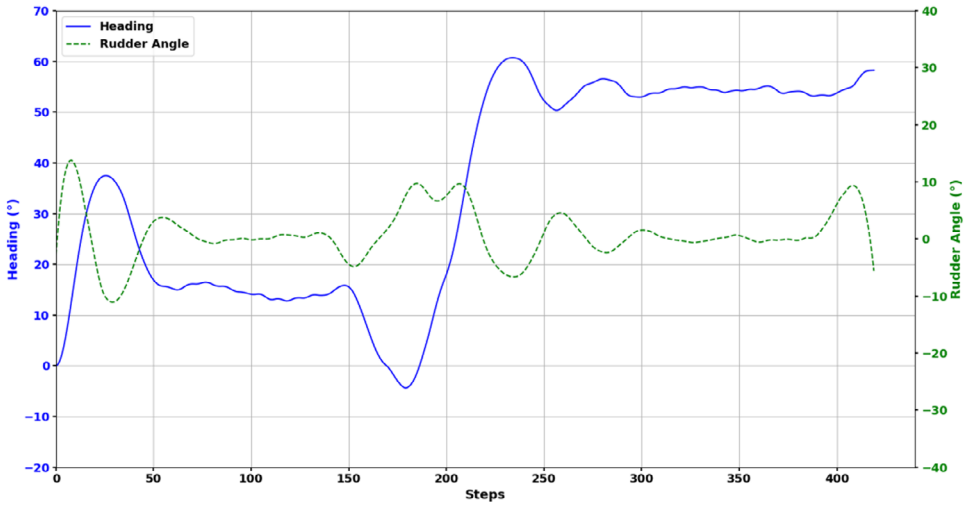


(a) Heading angle comparison between improved and traditional A*



(b) Improved heading angle and rudder angle

FIGURE 9 | Heading and ruder angle of scenario 2.

TABLE 6 | Performance metrics.

No.	Path length (m)	Final docking error (m)	Min clearance (m)	Avg curvature	RMS XTE (m)	Max XTE (m)
1	4630.74	80.46	57.48	0.01088	10.89	26.47
2	4403.06	86.53	57.17	0.01099	9.93	23.96
3	4685.91	85.24	56.48	0.01074	11.24	25.01
4	4704.12	79.88	58.72	0.01097	9.65	22.81
5	4392.82	81.05	58.14	0.01121	10.02	24.05

As shown in Figure 12, the path avoided obstacles represented by deep-blue zones and repulsive peak areas, maintaining compact and smooth detours in densely distributed obstacle regions (e.g., $X \in [2500, 4000]$, $Y \in [2000, 3500]$). Within the repulsive potential field, the path consistently remained in low-repulsion regions and the deep-blue potential field peaks did not intersect with the trajectory. These results demonstrated that APF effectively regulated the ship's trajectory in locally dynamic environments without requiring global re-planning. Overall, the 3D potential field visualisation and projection analysis validated the integration of APF into trajectory tracking as an effective approach to improving path smoothness and navigational safety.

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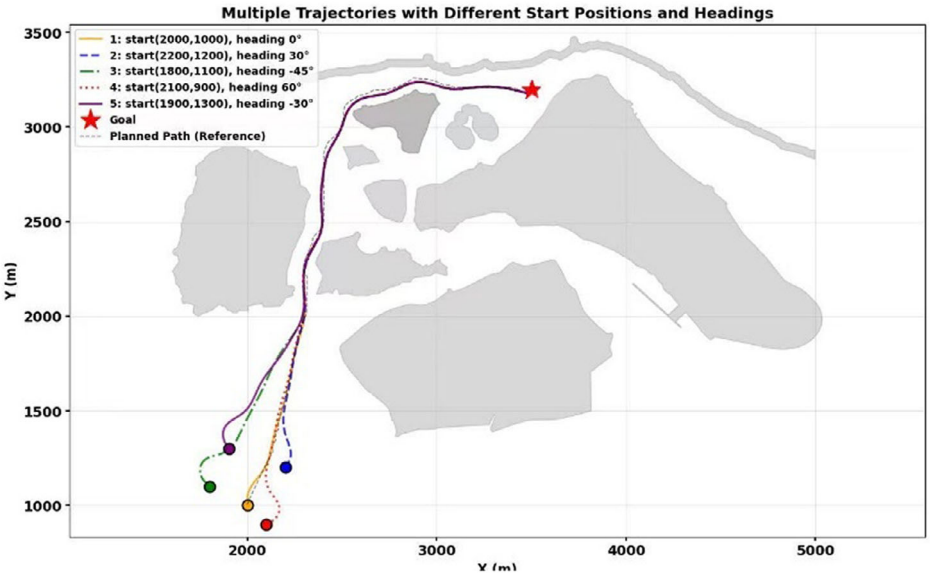
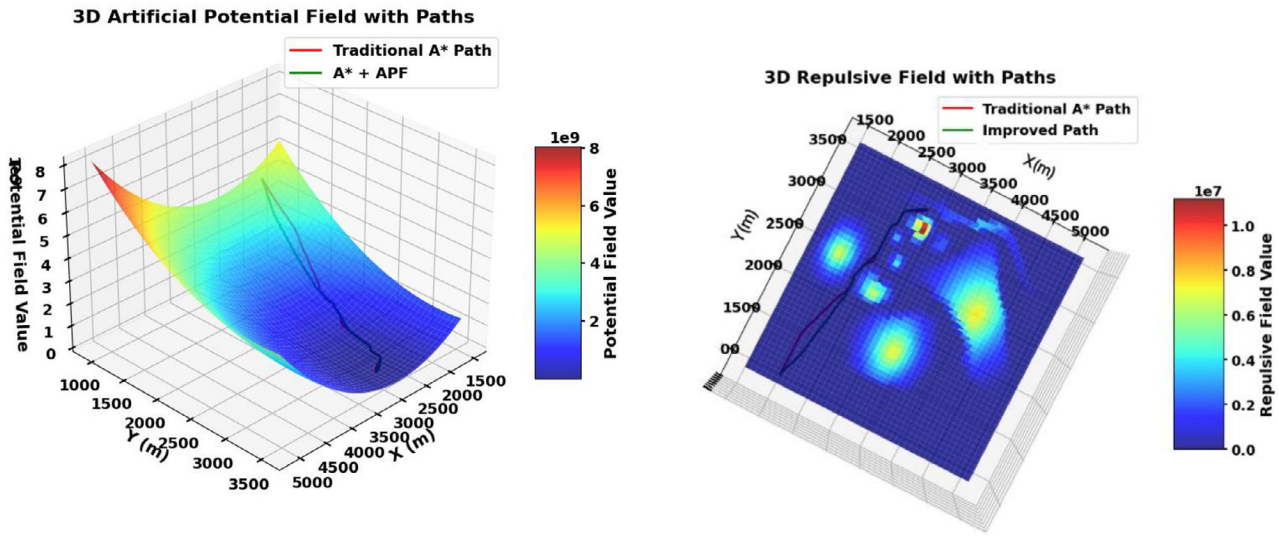


FIGURE 10 | Trajectories generated from different starting positions and headings.



(a) 3D Artificial Potential Field with Paths (b) Repulsion plane diagram

FIGURE 11 | Repulsive potential field visualisation.

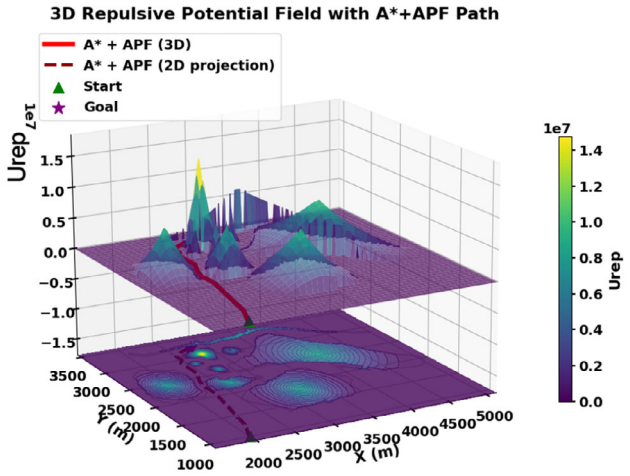


FIGURE 12 | 3D repulsive potential field with A* + APF path.

4.3 | Managerial Insights

The proposed hybrid A*-APF–MMG approach provides several managerial insights for port authorities and ship operators.

First, the method enables autonomous berthing planning based on real geographic data, reducing human dependence and improving safety in confined waters.

Second, by quantitatively evaluating path smoothness and obstacle clearance, the model supports data-driven decision-making for berth scheduling and tug assistance.

Third, the framework can be used as a digital tool for port risk assessment and pilot training, helping to design safer navigation routes and evaluate different berth layouts.

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Finally, integrating real-time environmental data (e.g., tides, wind and current) into future implementations could further enhance port automation and intelligent traffic management.

5 | Conclusion

Path planning and path tracking in berthing waters represent critical phases of a ship's voyage, requiring high precision in trajectory generation and manoeuvre control. To address the challenges posed by constrained manoeuvrability and complex obstacle environments, this paper proposed a two-stage hybrid method that integrates improved A* path planning with APF-based local refinement and dynamic path tracking control.

The proposed method addresses the limitations of traditional A* algorithms — such as non-smooth paths and poor adaptability to complex environments — and the inherent weaknesses of APF, such as susceptibility to local minima. The improved A* algorithm introduces a heading-change penalty to promote smoother and more navigable global paths, while the APF provides local refinements for obstacle avoidance and enhanced continuity. The final trajectory is smoothed using cubic spline interpolation and executed through a dynamic PID-based path tracking controller, which jointly adjusts rudder angle and propulsion speed to achieve accurate and stable tracking.

Simulation results in both virtual and real-world scenarios (including a real map of Zhoushan Port) demonstrate that the proposed method significantly improves trajectory smoothness, obstacle clearance and tracking performance. Specifically, in a regular obstacle environment, the minimum distance to obstacles increased from 15.12 m to 39.04 m, while the average turning angle decreased from 4.17° to 2.51°. In real-world tests, the heading deviation was effectively controlled within ±10° and the ship achieved smooth deceleration and accurate docking. These results confirm the method's practical applicability for real-time autonomous ship navigation in a constrained port environment. This study focused on static obstacle scenarios within confined berthing waters. Future research will focus on the following areas. First, to address the challenges of collision avoidance with moving obstacles, reliance on experience-based parameter settings and the computational burden caused by high-resolution geographic modelling in complex environments, future research will introduce an optimisation-based adaptive parameter tuning method. This method aims to improve control accuracy while maintaining computational efficiency in dynamic and high-complexity scenarios. Second, to better account for environmental influences and dynamic targets, a dynamic environmental perception model will be developed. This model will consider factors such as wind, waves and currents. In addition, a multi-ship interaction mechanism will be introduced to enable collision avoidance in the presence of moving ships. Finally, the method can be applied to real ships or higher-fidelity simulation platforms to validate its feasibility and scalability in real-world operations when entering berthing waters. The proposed framework can be readily deployed on hardware-in-the-loop simulators or scale-model platforms, serving as a validation step toward full-scale autonomous berthing trials.

Finally, although the present work is simulation-based, the proposed method is readily compatible with real-world deployment. The A–APF planning framework and the MMG-based tracking controller require only standard navigation measurements (position, heading and speed), making them suitable for integration into hardware-in-the-loop simulators or scaled physical ship platforms. In future work, the method will be validated on such platforms to assess robustness under realistic disturbances, providing an essential step toward full-scale autonomous berthing trials.

Author Contributions

Zhuo Wen: writing – review and editing, writing – original draft, visualisation, software, methodology, data curation, conceptualisation. **Jinfen Zhang:** writing – review and editing, writing – original draft, validation, supervision, resources, project administration, funding acquisition, formal analysis, data curation, conceptualisation. **Jiongjiong Liu:** writing – review and editing, writing – original draft, supervision, resources, project administration, methodology, funding acquisition, conceptualisation. **Wu Ning:** validation, resources, project administration, formal analysis, data curation.

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Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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